# Неопределени интеграли – Рационални функции

**Задача 1**

$$\int \frac{\mathrm{d}x}{(x-2)^3} \qquad \int \frac{\mathrm{d}x}{(7-2x)^5} \qquad \int \frac{\mathrm{d}x}{3x-5}$$

**Задача 2**

$$\int \frac{x^2\,\mathrm{d}x}{(1+x^2)^2} \qquad \int \frac{\mathrm{d}x}{(1+x^2)^3} \qquad \int \frac{\mathrm{d}x}{(4+x^2)^2}$$

**Задача 3**

$$\int \frac{\mathrm{d}x}{x^2-4x+13} \qquad \int \frac{3x+1}{x^2-4x+8}\,\mathrm{d}x$$

$$\int \frac{2x+2}{x^2+2x+17}\,\mathrm{d}x$$

$$\int \frac{\mathrm{d}x}{(x^2+6x+13)^2} \qquad \int \frac{\mathrm{d}x}{(x^2+2x+10)^2}$$

**Задача 4**

$$\int \frac{\mathrm{d}x}{x^2-x-2} \qquad \int \frac{(x^2+1)\,\mathrm{d}x}{(x^2-1)(x^2-4)}$$

$$\int \frac{x}{2x^2-3x-2}\,\mathrm{d}x$$

$$\int \frac{\mathrm{d}x}{x^3+4x^2+x-6} \qquad {}^{*}\int \frac{\mathrm{d}x}{x^4+1}$$

**Задача 5**

$$\int \frac{x^2-5x+9}{x^2-5x+6}\,\mathrm{d}x \qquad \int \frac{x^2\,\mathrm{d}x}{x^2-6x+10}$$

$$\int \frac{3x^3-5x+8}{x^2-4}\,\mathrm{d}x \qquad \int \frac{x^5+x^4-8}{x^3-4x}\,\mathrm{d}x$$

**Задача 6**

$$\int \frac{\mathrm{d}x}{(1+x^3)^2} \qquad \int \frac{x^4+2x^2+4}{(x^2+1)^3}\,\mathrm{d}x$$

$$\int \frac{(3x^2-2)x\,\mathrm{d}x}{(x+2)^2(3x^2-2x+4)} \qquad \int \frac{(x^7+2)\,\mathrm{d}x}{x^4+2x^3+3x^2+2x+1}$$